

BAIA MARE TAUTII MAGHERAUS, Romania

WMO: 150140

Lat: 47.6608N

Lon: 23.4733E

Elev: 187

StdP: 99.09

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.0	-9.9	-15.8	1.0	-11.4	-12.8	1.3	-8.4	7.1	2.3	6.0	1.4	1.3	90	0.472

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.4	32.3	21.4	30.9	21.0	29.1	20.0	22.9	30.1	21.9	28.5	21.0	27.4	2.4	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.8	15.8	26.1	19.8	14.8	25.1	18.9	14.0	24.0	68.8	29.9	65.0	28.6	61.5	27.5	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.9	5.8	5.0	-17.3	34.6	4.0	1.8	-20.1	35.8	-22.4	36.9	-24.7	37.9	-27.5	39.2		
(5)			-17.7	24.9	4.0	1.3	-20.5	25.9	-22.8	26.7	-25.1	27.4	-28.0	28.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	-1.1	1.3	5.6	11.6	16.2	19.8	21.2	21.4	16.2	11.0	5.9	0.6
	DBStd	8.93	4.85	5.07	4.50	4.15	3.61	3.41	3.17	3.21	3.90	4.28	4.76	4.47
	HDD10.0	1241	344	245	146	31	2	0	0	0	2	41	138	292
	HDD18.3	3042	602	478	396	203	84	24	10	9	86	229	372	550
	CDD10.0	1554	0	1	9	78	196	294	348	353	188	72	16	0
	CDD18.3	314	0	0	0	2	19	68	99	103	22	1	0	0
	CDH23.3	3104	0	0	0	26	211	642	949	1027	237	13	0	0
	CDH26.7	1058	0	0	0	3	37	196	352	407	62	1	0	0
(14) Wind	WSAvg	2.1	1.7	2.0	2.4	2.5	2.3	2.3	2.2	2.2	2.1	1.9	1.9	1.8
(15) Precipitation	PrecAvg	816	59	51	52	55	76	96	87	58	77	72	66	74
	PrecMax	1152	107	128	177	95	186	180	182	108	164	171	128	138
	PrecMin	565	24	20	13	14	28	14	15	9	19	11	2	6
	PrecStd	141	18	25	40	21	38	41	51	27	39	43	34	37
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.8	14.9	19.9	26.2	29.8	32.2	34.9	35.1	31.6	25.0	19.8	12.2
		MCWB	8.4	9.5	12.1	15.8	19.4	21.8	22.3	21.4	19.2	16.2	13.3	8.6
	2%	DB	8.6	11.9	17.1	23.2	27.8	30.8	32.1	32.9	28.1	22.9	17.0	10.0
		MCWB	7.0	7.7	10.3	14.2	18.2	21.3	22.2	21.0	17.7	15.6	12.0	7.9
	5%	DB	6.9	9.4	14.9	21.4	25.9	29.0	30.2	31.0	26.1	20.2	15.0	8.1
		MCWB	5.6	6.7	9.3	13.2	17.5	19.9	21.2	20.8	17.6	14.7	11.2	6.7
	10%	DB	4.9	7.6	12.3	19.3	23.9	27.1	28.7	28.9	23.8	17.9	12.9	6.3
		MCWB	4.0	5.8	8.0	12.3	16.5	19.5	20.5	20.2	17.0	13.8	10.1	5.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.0	9.9	12.8	16.5	21.1	23.9	24.2	24.0	20.7	17.7	14.4	9.9
		MCDB	10.3	13.9	18.0	24.0	27.6	30.1	30.8	30.8	27.6	22.2	17.3	11.2
	2%	WB	7.3	8.3	11.1	15.3	19.3	22.4	22.9	22.6	19.3	16.4	13.0	8.0
		MCDB	8.1	10.7	15.7	21.8	25.2	28.6	30.5	29.7	25.4	20.6	16.2	9.4
	5%	WB	5.6	7.0	9.9	14.1	18.0	21.2	21.9	21.5	18.3	15.3	11.7	6.8
		MCDB	6.5	8.9	13.6	19.5	23.7	27.0	28.9	28.3	24.1	19.3	14.5	7.8
	10%	WB	4.1	5.8	8.6	12.9	17.0	20.0	21.0	20.6	17.4	14.2	10.4	5.5
		MCDB	4.8	7.4	11.9	17.8	22.6	25.5	27.3	27.2	22.7	17.5	12.5	6.3
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	7.0	9.3	11.0	11.3	11.4	11.7	12.4	11.0	10.4	7.7	5.1
		MCDBR	7.0	9.5	13.2	14.1	14.6	13.7	14.3	15.1	14.7	13.7	10.7	7.6
		MCWBR	5.3	6.3	7.5	6.8	6.8	6.1	6.2	5.6	6.3	7.3	6.6	5.5
	5% WB	MCDBR	6.2	8.5	11.5	12.3	13.0	12.4	13.5	13.3	12.9	11.7	9.3	6.8
		MCWBR	5.0	5.9	6.9	6.4	6.6	6.2	6.3	5.8	6.2	6.7	5.9	5.3
(40) Clear-Sky Solar Irradiance	taub		0.329	0.358	0.403	0.457	0.439	0.436	0.458	0.458	0.418	0.386	0.358	0.330
	taud		2.246	2.195	2.149	2.053	2.171	2.218	2.168	2.163	2.232	2.288	2.304	2.296
	Ebn at Noon		722	782	802	794	827	831	805	785	780	743	684	674
	Edh at Noon		80	104	128	155	143	137	142	137	117	95	76	68
(44) All-Sky Solar Radiation	RadAvg		1.02	1.84	3.07	4.44	5.43	5.91	5.80	5.33	3.70	2.37	1.29	0.76
	RadStd		0.16	0.28	0.46	0.47	0.56	0.39	0.51	0.48	0.46	0.31	0.19	0.12

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (47 neighbors)	+1.01	N/A	N/A	+1.42	N/A	N/A	N/A	N/A	N/A	N/A
		+0.70	N/A	N/A	+1.08	N/A	N/A	-136	-209	+120	+55

Nomenclature: See separate page